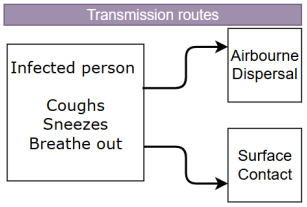
Introduction



Context and Disease Information : Measles virus is highly contagious, disproportionately affecting children. The virus spreads via airborne transmission when an infected person breathes, coughs or sneezes. The virus can remain active in the air or on surfaces for up to two hours, increasing transmissibility via indirect contact.

First symptoms (prodromes) are non-specific similar to a cold, e.g. high temperature, blocked or runny nose, sneezing, coughing, sore watery eyes. Four days later (on average) a red rash appears with spots sometimes raised or forming blotchy patches. A person is typically infectious from symptom onset until four days after the rash appears. An episode of measles confers lifelong immunity. Measles can spread to the lungs and brain, complications can be serious and sometimes life-changing, including pneumonia, meningitis, blindness, seizures, death. [3]

Research question : How well will applying the differential equation modelling methods taught in the course allow the Hagelloch measles outbreak data from a small, contained outbreak in 1861 to be mathematically modelled ?

Data summary

Dataset #1 - A measles dataset found in the R surveillance package - Measles in Hagelloch, Germany (1861). Hagelloch is a small secluded German town providing a well documented example of an outbreak in a contained population. Dr. Albert Pfeilsticker made daily records at the time of the outbreak for the 188 cases [1], Dr. Heike Oesterle (1992) augmented and re-analysed the dataset [2]. The data consists of a very detailed patient based recordset with dates of symptom onset, rash onset, recovery, complications etc. Contact tracing of infector and infectee was observed, so the data contains (with reasonable certainty) the transmission network.

Schema for Hagelloch dataset
Infected Individuals

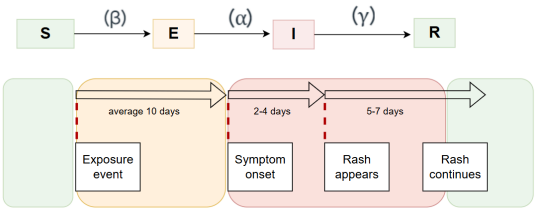
Variable	description	order	datatype	count_unique	count_na
PN	patient number	1	integer	187	0
IFTO	patient no infection source	2	integer	62	3
AGE	age in years	3	numeric	20	0
PRO	Date of prodromes	4	Date	35	0
tPRO	first symptoms days post (30 Oct 1861)	5	numeric	187	0
ERU	Date of rash	6	Date	34	0
tR	Day infectious ends	7	numeric	187	0
GE	generation number of case	8	integer	2	0
CL	school class	9	factor	3	0
x.loc	x coord (in meters)	10	numeric	41	0
y.loc	y coord (in meters)	11	numeric	44	0
days_infected	Number of days infected (calculated)	12	numeric	12	0
tINFECTING_PRO	Prodromes of infecting patient (calculated)	13	numeric	62	3
SI	Serial Interval (calculated)	14	numeric	184	3

Dataset #2 - Annual Measles Notifications and Deaths in England and Wales, 1940 to 2025 published by gov.uk [4].

Schema for England and Wales Measles dataset
1940 - 2025

Variable	order	datatype	count_unique	count_na
Year	1	numeric	86	0
Notifications	2	numeric	86	0
Total deaths	3	numeric	50	0

Methods



Model formulation : The compartmental model for this disease includes an Exposed compartment to capture the long incubation period of the disease.

Compartment Equations - Formulae

$$\Delta S = -\frac{\beta SI}{N}$$

$$\Delta E = \frac{\beta SI}{N} - \alpha E$$

$$\Delta I = \alpha E - \gamma I$$

$$\Delta R = \gamma I$$

Compartment Equations - Parameters

- Transmission Rate (β): Rate at which an infected person infects others.
- Incubation Rate (α): Rate at which exposed individuals become infectious, calculated as 1/d1, where d1 is the average number of days exposed before symptoms develop.
- Removal Rate (γ): Rate at which individuals are removed or recovered from the infectious pool, calculated as 1/d2, where d2 is the number of days to recover.

Assumptions : For this outbreak there were no systemic interventions

Methods : As the Hagelloch outbreak contains contact tracing data, and dates of symptoms, recovery, we can calculate SI (Serial Interval), and SEIR compartments representing the progress of the disease for each individual. These can be used as context for the model outcome.

Results

Results - understanding what the data tells us about the progress of the outbreak

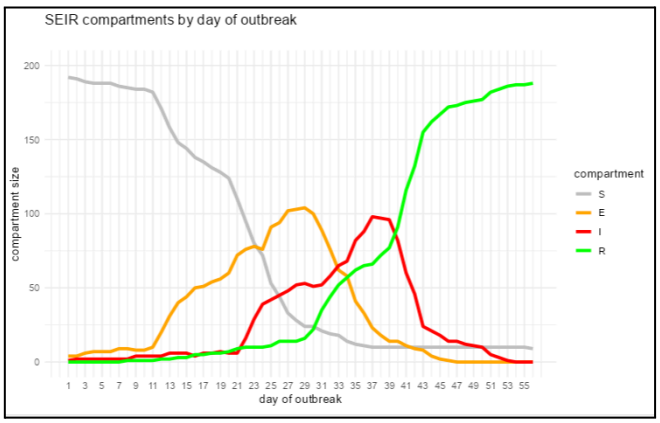


Chart 1. Calculate SEIR compartments from the data.

There were 2 generations of infection, shown clearly in this chart. The first wave of infections is waning and may actually be beginning to decline as the second rises steeply.

The variable GE records this.

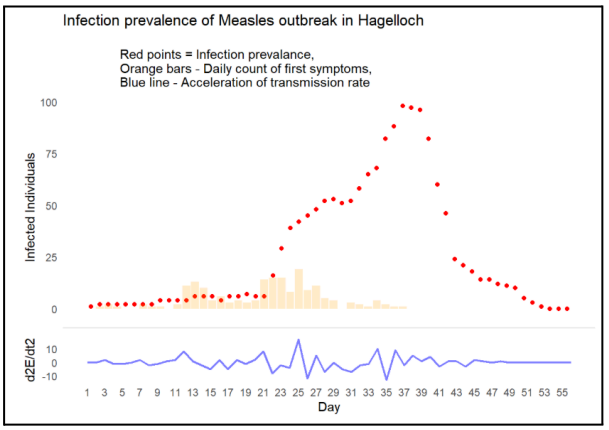


Chart 2. Show how Infection prevalence follows symptom onset, calculate acceleration of the transmission by taking the second derivative of the prevalence values.

The chart shows how the long incubation period allows the virus to spread in the population before symptoms emerge. By the time the Infection prevalence begins to rise, a high proportion of the population is already exposed.

Results - checking parameter assumptions for the model

Population Statistics for Wave 1 and 2 of Hagelloch Measles Outbreak
Infected Individuals only

description	wave_1	wave_2
Total sample	109.0	78.0
mean_SI	10.1	10.7
mean_Infected_days	7.4	6.4
infected_on_day1	1.0	0.0
infected_on_day9	4.0	0.0

Table 1. Calculating values for -

- incubation α (10 days from mean_SI)
- removal γ (7 days from mean_Infected_days).

For the initial compartment levels we know the number infected on day 1. There were 4 infected individuals on day 9, on the day 1 these individuals must have been exposed. From the literature we know there were 192 Susceptible individuals, so these are the starting values for the model.

Results - modelling the outbreak

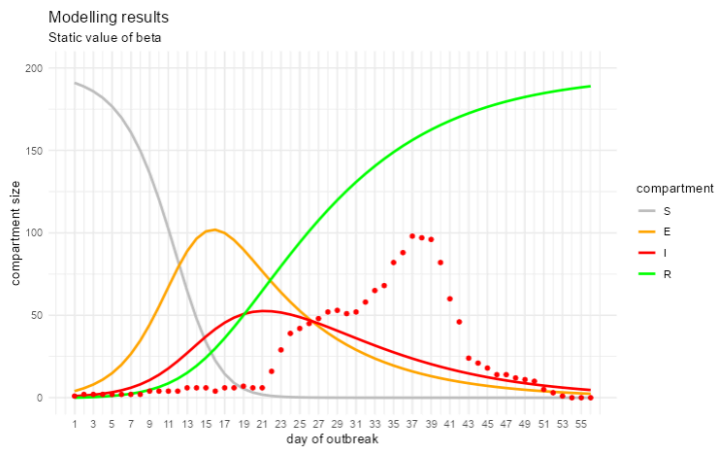


Chart 3. Modelling the SEIR equations using a static value for beta.

The Susceptible compartment is quickly exhausted, and the modeled Infection compartment peaks much earlier than the observed data.

Hypothesis : There may be a variable transmission rate for some reason, probably changes to the pattern of social contacts.

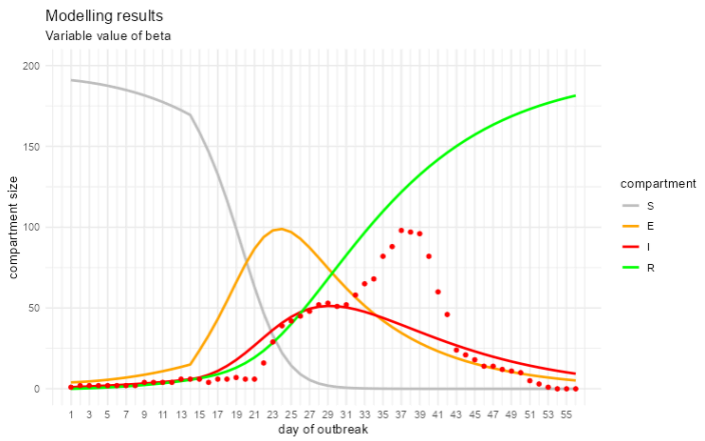


Chart 4. Modelling with a varying beta over time.

This allows a slower movement from S to E, and the first wave is modeled fairly accurately, but there are no susceptible individuals left to feed the second wave.

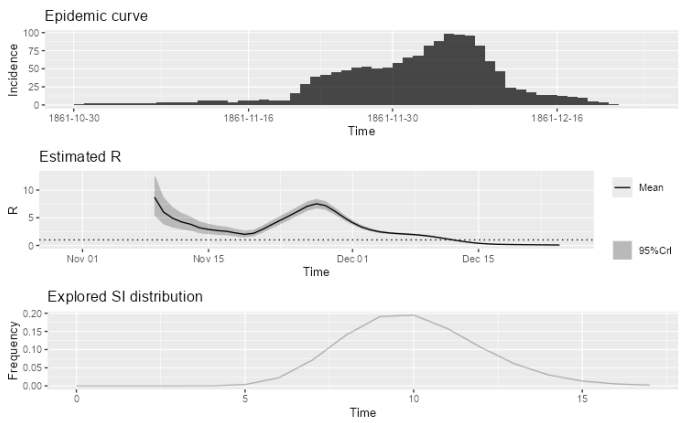
At this point we're straying into the territory of overfitting with such a small sample.

Results - look for other evidence of a variable transmission rate

Find the value of R (reproduction number) using the EpiEstim package

Chart 5. EpiEstim plots show how the value of R varies over the course of the outbreak.

As R = beta / gamma, and we know from the data and the medical knowledge that gamma is fairly constant, the variation must come from changing value of beta, confirming the SEIR modelling hypothesis.



Results - looking at the contact tracing network

Table 2. Looking at the number of secondary infections each infected individual caused, we can see that there were a few significant 'super spreader' individuals. Patient number 45 is recorded as infecting 30 individuals.

This explains the variable transmission rate, and also shows how it's not possible to build a generalisable SEIR model from this small dataset without overfitting or data leakage. The value of this dataset is in understanding measles transmission in a small social network, which is likely to be the precursor to larger outbreaks.

Contact Network
Highest secondary infections

Source Individual	Number Infected	Date First Symptoms	Day First Symptoms
45	30	1861-11-11	13
181	8	1861-11-13	14
7	7	1861-11-24	25
17	7	1861-11-20	22
180	7	1861-11-17	18
153	6	1861-11-24	26

See the animation in the supplied markdown code showing disease progression geographically through the contact network over time. Using the gganimate package and the x, y coordinates of homes in Hagelloch.

Conclusion : Given the highly inconsistent transmission rate it's not possible to build a generalisable model from the Hagelloch data. The data does confirm the exposure and recovery rates most frequently quoted in the literature.

Discussion - conclusions and their relevance to policy

The Hagelloch outbreak demonstrates why measles is such a dangerous disease. 12 of the 187 (6.5%) infected died and 37 (19.7%) individuals developed severe complications. This is in the context of a rural town in 1861, doubtless modern medical care and hygiene standards are superior, and nutrition is improved. However the fundamental characteristics of easy transmission, long incubation period and potentially severe complications mean Measles has a high human and economic impact, although is rarely fatal these days.

The dataset shows how the virus can spread 'silently' for a significant period, when first symptoms emerge many contacts will already be exposed. The ability of the virus to live undetected on surfaces for a significant period of time adds to transmissibility.

Policy interventions such as quarantines or lockdowns could reduce the spread of Measles if they were introduced early enough in the outbreak. Given the 10 day incubation period, however, the quarantine period would need to be widespread and lengthy. Contact tracing has similar drawbacks, by the time symptoms emerge the network of exposed contacts for the previous 10 days is likely to be large, especially during term time.

The only intervention offering a realistic firebreak is a vaccination program (assuming a safe vaccine). As Measles confers life long immunity this can be a single dose in a person's life, therefore cost effective and administratively manageable. The vaccination program needs to be implemented as early in a child's life as medically safe. The Hagelloch data documents 7 infants under 6 months old not infected, hypothesising protection from maternal antibodies.

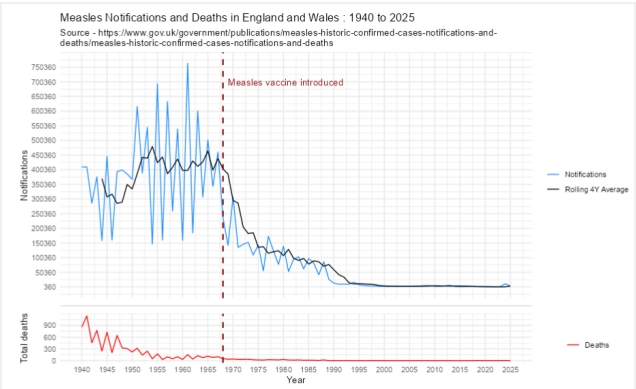


Chart 6. Data for Measles Notifications and Deaths in England and Wales from 1940 onwards is open source.

Deaths from Measles were falling from 1940, a hypothesis would be that multiple conditions such as improving medical care and lower birth rate due to WW1 and WW2 were factors. From 1950 notifications rose significantly, possibly due to increased birth rate. The 4 year rolling average shows around 400,000 infections per year, with periodic outbreaks spiking higher. Happily, the death rate continued to fall during this time.

Measles can lead to serious complications with life-long repercussions, pneumonia, blindness, deafness to name a few. The strain on both families and the Health Service is significant. There's also the possibility that the virus might mutate into a more deadly form if it were to be uncontrolled.

The introduction of a vaccine in 1968 had an immediate and dramatic effect on Measles notifications. Although not eradicated, these days Measles is fairly rare in the UK with a few annual, usually localised, isolated outbreaks,

The immunisation program is still needed to maintain a level of herd immunity. The willingness of parents to present their children for immunisation relies on policy makers maintaining a high degree of public trust and confidence that vaccines are both safe and effective.

Personal Note

I've enjoyed this course enormously. It's been super-interesting to understand the compartmental modelling approach, which I have realised can be applied to many other domains. For instance we could build trust models of vaccination programs and examine how negative publicity affects public uptake. Similarly a brand advertising campaign might aim to move individuals from a not-aware to an aware compartment.

References

[1]	http://www.archive.org/details/beitrgezurpatho00pfeigoog Author : Pfeilsticker, A. Title : Beiträge zur Pathologie der Masern mit besonderer Berücksichtigung Published : 1863
[2]	https://www.genealogy.math.ndsu.nodak.edu/id.php?id=284204 Author : Dr Heike Oesterle Title : Statistische Reanalyse einer Masernepidemie 1861 in Hagelloch Published : 1992
[3]	https://www.cdc.gov/measles/signs-symptoms/index.html Author : CDC (Centre for Disease Control) Title : Measles Symptoms and Complications Published : 9th May 2024
[4]	https://www.gov.uk/government/publications/measles-historic-confirmed-cases-notifications-and-deaths/measles-historic-confirmed-cases-notifications-and-deaths#measles-notifications-and-deaths-in-england-and-wales-1940-to-2025 Author : gov.uk Title : Measles notifications and deaths in England and Wales: 1940 to 2025 Published : 8th January 2026